

Problem 1

Let M and N be normal subgroups of a group G . prove that MN is a normal subgroup of G

Solution:

First we should show that $MN \leq G$

Note that $MN \neq \emptyset : 1 \cdot 1 = 1 \in MN$

Additionally for $m_1, m_2 \in M$ and $n_1, n_2 \in N$ we can show that closure and inverses hold:

Closure: $(m_1 n_1)(m_2 n_2) = m_1 n_1 m_2 n_2 = m_1 (n_1 m_2 n_1^{-1}) n_1 n_2$ and $n_1 m_2 n_1^{-1} \in M : M \trianglelefteq G \Rightarrow$ closure holds

Inverse: note that for any $mn \in MN$ the inverse (both left and right) is $(mn)^{-1} = n^{-1}m^{-1}$, and this inverse is in MN since $n^{-1}m^{-1} = (n^{-1}m^{-1}n)n^{-1}$ and $n^{-1}m^{-1}n \in M : M \trianglelefteq G \Rightarrow n^{-1}m^{-1} \in MN$

To prove that $MN \trianglelefteq G$ would like to show that $gmn g^{-1} \in MN, \forall g \in G, \forall m \in M, \forall n \in N$

Note that $gmn g^{-1} = gm g^{-1} g n g^{-1} = (gm g^{-1})(g n g^{-1}) \in MN$ since by assumption $gm g^{-1} \in M$ and $(g n g^{-1}) \in N$

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Problem 2

Let M and N be normal subgroups of G such that $G = MN$. Prove that $G/(M \cap N) \cong (G/M) \times (G/N)$

Solution:

We can define a map $\varphi : G \rightarrow (G/M) \times (G/N)$ by $g \mapsto gM \times gN$

Well defined: suppose $g_1 = g_2$ for $g_1, g_2 \in G$

Then $\varphi(g_1) = (g_1 M, g_1 N)$ and $\varphi(g_2) = (g_2 M, g_2 N)$

Then $\varphi(g_1) = \varphi(g_2)$ by the assumption that $g_1 = g_2$

Homomorphism: $\varphi(g_1 g_2) = (g_1 g_2 M, g_1 g_2 N) = (g_1 M g_2 M, g_1 N g_2 N) = (g_1 M, g_1 N)(g_2 M, g_2 N) = \varphi(g_1)\varphi(g_2)$

Surjective: let $g_1, g_2 \in G$ where $g_1 = m_1 n_1, g_2 = m_2 n_2$ then we would like to show that $\varphi(m_2 n_1) = (g_1 M, g_2 N)$

This is true because $g_1 M = m_1 n_1 M = n_1 n_1^{-1} m_1 n_1 M = n_1 M$ and that

$g_2 N = m_2 n_2 N = m_2 N$

Finally, $\varphi(m_2 n_1) = (m_2 n_1 M, m_2 n_1 N) = (n_1 n_1^{-1} m_2 n_1 M, m_2 n_1 N) = (n_1 M, m_2 M) = (g_1 M, g_2 N)$

The kernel of φ is all elements $g \in G$ such that $\varphi(g) = (gM, gN) = (M, N)$ which only happens if $g \in M \cap N$ (from a prop in class) so the kernel of φ is $M \cap N$

Applying the first isomorphism theorem, the claim follows

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reference:

<https://math.stackexchange.com/questions/310351>

Problem 3

Let p be a prime and let G be a group of order $p^a m$ where p does not divide m . Assume P is a subgroup of G of order p^a and N is a normal subgroup of G of order $p^b n$ where p does not divide n . Prove that $|P \cap N| = p^b$ and $|PN/N| = p^{a-b}$. (the subgroup P of G is called a Sylow p -subgroup of G . This exercise shows that the intersection of any Sylow p -subgroup with a normal subgroup N is a Sylow p -subgroup of N .)

Solution:

By the second isomorphism theorem we have that

$$(PN)/N \cong P/(P \cap N) \Rightarrow |(PN)/N| = |P/(P \cap N)| = |PN| / |N| = |P| / |P \cap N|$$

$$\Rightarrow |PN| = \frac{|P| |N|}{|P \cap N|} \Rightarrow |PN| = \frac{p^a p^b n}{p^{a-b}}$$

Thus we would like to show that $|PN| = p^a n$ then the claim will follow

By Lagrange's we know that there exist k such that $|PN| \cdot k = p^a m$ since $PN \leq G$

Then equivalently we would like to show that $p^a m = p^a n k \Rightarrow m = nk$

Suppose that $m \neq nk$ then $n \nmid m$

From Lagrange's we know that $p^b n \mid p^a m \Rightarrow n \mid m$ which invalidates this assumption, and the claim follows

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Problem 4

Show that if $|G| = pq$ for some primes p, q (not necessarily distinct) then either G is abelian or $Z(G) = \{1\}$

Solution: Problem 3, Homework 7

by Lagrange's theorem, the possible orders of $Z(G)$ are $1, p, q, pq$

Suppose that $|Z(G)| = 1$ then $Z(G) = \{1\}$ (trivial group is a group)

Suppose that $|Z(G)| = pq$ then $Z(G) = G$ so that G is abelian

Suppose that $|Z(G)| = p$ then $|G/Z(G)| = pq/p = q$ (from class) since q is prime $G/Z(G)$ is cyclic (from previous homework). This makes G abelian (from problem 2 (see below)) i.e. that $Z(G) = G$ leading to contradiction $p = pq$ so we conclude $|Z(G)| \neq p$

Similarly, $|Z(G)| \neq q$

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For some $x \in G$ let the generator of $G/Z(G)$ be $xZ(G)$ then we have that for $g \in G$ that $gZ(G) = (xZ(G))^a$ for some a and using the binary operation of $G/Z(G)$ we have that $(xZ(G))^a = x^a Z(G)$

Since we have that $gZ(G) = x^a Z(G)$ then by a proposition in class $(x^a)^{-1}g \in Z(G)$, so let $x^{-a}g = z$ for some $z \in Z$ then we have $g = x^a z$

Since z is from $Z(G)$ and commutes with all elements of G , let $g_1 = x^{a_1} z_1$ and $g_2 = x^{a_2} z_2$ then

$$g_1 g_2 = x^{a_1} z_1 x^{a_2} z_2 = x^{a_2} z_2 x^{a_1} z_1 = g_2 g_1$$

where $g_1, g_2 \in G$ and so G is abelian

Problem 5

Show that the additive group $\mathbb{R}$ acts on the x, y plane $\mathbb{R} \times \mathbb{R}$ by $r \cdot (x, y) = (x + ry, y)$

Solution:

We can check the two properties of a left group action: let $r_1, r_2 \in (\mathbb{R}, +)$ and $(x, y) \in \mathbb{R} \times \mathbb{R}$ then

$$1. (r_1 + r_2) \cdot (x, y) = (x + (r_1 + r_2)y, y) = (x + r_1y + r_2y, y)$$

$$r_1 \cdot (r_2 \cdot (x, y)) = r_1 \cdot (x + r_2y, y) = (x + r_2y + r_1y, y) = (x + r_1y + r_2y, y) \text{ since addition is commutative in } \mathbb{R}$$

$$2. 0 \cdot (x, y) = (x + 0y, y) = (x, y)$$

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Problem 6

For each $n \geq 1$ find all normal subgroups of S_n

Solution:

$S_1 = \{1\}$ and has normal subgroups $\{1\}$ and S_1

$S_2 = \{(), (1\ 2)\}$ and also has normal subgroups $\{1\}$ and S_2

$S_3 = \{(), (1\ 2), (2\ 3), (1\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Besides $\{1\}$ and S_3 it also has normal subgroup A_3 since it is the kernel of the sign homomorphism $f : S_n \rightarrow \{1, -1\}^*$ where $\{1, -1\}^* \cong \mathbb{Z}_2$ due to the isomorphism $f(x) = \frac{1}{2}(1 - x)$

None of the order 2 subgroups generated by a single transposition are normal since conjugation of a transposition by a transposition will result in a different transposition, for example $(1\ 3)(1\ 2)(1\ 3) = (2\ 3)$

S_4 has normal subgroups $\{()\}$, S_4 , A_4 and also $V_4 = \{(), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ as we saw in class

For $n \geq 5$ we have S_n , $\{()\}$, and A_n

Since A_n is simple for $n \geq 5$ then its only normal subgroups are trivial subgroup and itself

For a proof that A_n is the only normal subgroup besides S_n and $\{()\}$ we can check the following post:

<https://math.stackexchange.com/questions/482889/normal-subgroups-of-s-n-for-n-geq-5>

Problem 7

Assume n is an even positive integer and show that D_{2n} acts on the set consisting of pairs of opposite vertices of a regular n -gon. Find the kernel of this action

Solution:

Let $D_{2n} = \{1, r, r^2, \dots, r^{n-1}, s, sr, \dots, sr^{n-1}\}$ be the usual Dihedral group of order $2n$

We would like to define a group action $\mu : D_{2n} \times A \rightarrow A$ where $A = \{(i, i + \frac{n}{2}) : 0 \leq i \leq \frac{n}{2} - 1\}$ such that μ satisfies the “identity” and “compatibility” axioms for a group action (here A represents the pairs of opposite vertices)

Define μ by the following :

(i) $(1, (i, i + \frac{n}{2})) \mapsto (i, i + \frac{n}{2})$ then μ immediately satisfies the identity axiom and the compatibility axiom is satisfied by $1 \cdot (i, i + \frac{n}{2}) = 1 \cdot (i, i + \frac{n}{2}) = (i, i + \frac{n}{2}) = 1 \cdot (1 \cdot (i, i + \frac{n}{2}))$

where 1 is the multiplication of $1 \in D_{2n}$ with itself

(ii) rotations are mapped by $(r^k, (i, i + \frac{n}{2})) \mapsto (i + k, i + k + \frac{n}{2})$ where each coordinate is taken mod n . Clearly, $(i + k, i + k + \frac{n}{2}) \in A$ and we have that $(r^{k_1} r^{k_2}) \cdot (i, i + \frac{n}{2}) = r^{k_1 + k_2} (i, i + \frac{n}{2}) = (i + k_1 + k_2, i + k_1 + k_2 + \frac{n}{2}) = r^{k_1} \cdot (i + k_2, i + k_2 + \frac{n}{2}) = r^{k_1} \cdot (r^{k_2} \cdot (i, i + \frac{n}{2}))$ so that this map satisfies compatibility

(iii) rotations that are not across a line of reflection across midpoints can be mapped by $(sr^k, (i, i + \frac{n}{2})) \mapsto (i + \frac{n}{2}, i + \frac{n}{2} + \frac{n}{2}) = (i + \frac{n}{2}, i)$ where coordinates are taken mod n clearly $(i + \frac{n}{2}, i)$ is in A since the order of coordinates in A does not matter, and we have that $(sr^k sr^k) \cdot (i, i + \frac{n}{2}) = 1 \cdot (i, i + \frac{n}{2}) = (i, i + \frac{n}{2})$ and on the other hand $sr^k \cdot (sr^k \cdot (i, i + \frac{n}{2})) = sr^k \cdot (i + \frac{n}{2}, i) = (i, i + \frac{n}{2})$ so that this mapping satisfies compatibility

(iv) rotations that are across midpoints don't have an explicit mapping, however since these reflections preserve midpoints between vertices (but will relabel the vertices), then the action of such a reflection on an element in A will still be in A

And similar to (iii) since reflections have order 2 then case (iv) will also satisfy compatibility

Additionally since combinations of a rotation and reflection (in either order) are still reflections, then μ is a group action

The kernel of μ is $\{1, r^{\frac{n}{2}}\}$ since these are the only elements of D_{2n} which preserve all opposite pairs of vertices

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Problem 8

Let G be a group and let $A = G$. Show that the maps defined by $g \cdot a = ag^{-1} \forall g, a \in G$ satisfy the axioms of a (left) group action of G on itself.

Solution:

Note that ag^{-1} is computed in G with G 's binary operation in the definition of the map

(i) $1 \cdot a = a(1^{-1}) = a$

(ii) $(gh) \cdot a = a(gh)^{-1} = ah^{-1}g^{-1} \in A$

$g \cdot (h \cdot a) = g \cdot (ah^{-1}) = ah^{-1}g^{-1} \in A$ where $ah^{-1} \in A$

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